Paul He

416 659 5646 | paul.he@mail.utoronto.ca | linkedin | GitHub | Academic Website

Education

University of Toronto — September 2021 - May 2025
Honours Bachelor of Science in Computer Science — *Toronto, Canada*

- Dean's List
- **Relevant Coursework:** Introduction to Machine Learning

Swiss Federal Institute of Technology (ETH Zürich) — September 2023 - August 2024
Exchange Year — *Zürich, Switzerland*

- EESTEC Generative AI hackathon winner
- **Relevant Coursework:** Natural Language Processing, Probabilistic Artificial Intelligence, Machine Perception, Large Language Models, Advanced Formal Language Theory

Experience

Student Researcher — June 2024 – August 2024
Supervisors: Tianyu Liu, Prof. Dr. Ryan Cotterell — *Rycolab, Institute of Machine Learning, ETH Zürich*

- Developed a likelihood-tuning method to **increase the accuracy of open language models by up to 3%**.
- Conducted experiments by implementing the proposed mechanism.
- Analyzed experimental results and created synthetic data to generalize problem scenarios.

Projects

Speeding up Earley's Algorithm with A* Search — April 2024 – August 2024

- Proposed and implemented a grid search approach, resulting in a faster Earley's algorithm.
- Tested and experimented with the proposed algorithms on Treebank datasets to validate the theory.
- Ensured code correctness by implementing tests and debugging issues.

Word Segmentation and Part-of-Speech Tagging with Transformer — October 2023 – January 2024

- Designed a **new decoder embedding mechanism**, allowing the decoder to better capture subtle linguistic nuances and dependencies between characters and tags.
- Implemented a **transformer** model using **PyTorch** with beam-search for generation.
- Achieved leading-edge results with validation accuracy of **96.29%** for CWS, **93.37%** for POS tagging, and **91.99%** for joint CWS and POS tagging after just 10 epochs.

Mental Health Support Bot | *Python, BeautifulSoup, FastAPI* — October 2023

- Led the development of a simple mental health chatbot web app tailored for ETH Computer Science students, assisting team members with debugging.
- Performed **web scraping** to obtain data for **few-shot prompting**.
- Implemented a real-time chat feature using FastAPI to host the chat platform and model on a server, allowing multiple simultaneous chats with the bot and recall of previous conversations.

Clinic Management System | *Java, JUnit, Git* — June 2022 – August 2022

- Developed a Clinic Management System adhering to **Clean Architecture** and **SOLID principles**, incorporating **Design Patterns** for optimal code structure.
- Collaborated with a **team of 8**, leveraging **Git & GitHub** for efficient version control, pull requests, issue tracking, and code reviews.
- Ensured code reliability through comprehensive test coverage by developing robust test suites using the JUnit framework.

Technical Skills

Programming Languages: Python, C, Java, R, SQL
Frameworks/Libraries: PyTorch, scikit-learn, NumPy, FastAPI, Flask
Tools/Platforms: Git, Docker, Jupyter, GitHub
Databases: PostgreSQL, psycopg2